8.3 Reach for the Stars (STA)

The focus of this unit is on the physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe:

- distances of stars
- masses of stars
- energy sources in stars
- star formation
- star death and the creation of chemical elements
- the history and future of the universe.

This topic uses the molecular kinetic theory model of matter and includes a study of the 'Big Bang' model of the universe. It also involves mathematical modelling of gravitational force and radioactive decay.

There are opportunities for students to develop ICT skills using the internet, data-logging and simulations.

There are several case studies that show how scientific knowledge and understanding have changed over time, providing students with opportunities to consider the provisional nature of scientific ideas.

Students will be assessed on their ability to:		Suggested experiments
109	investigate, recognise and use the expression $\Delta E = mc\Delta\theta$	Measure specific heat capacity of a solid and a liquid using, for example, temperature sensor and data logger
110	explain the concept of internal energy as the random distribution of potential and kinetic energy amongst molecules	
111	explain the concept of absolute zero and how the average kinetic energy of molecules is related to the absolute temperature	
112	recognise and use the expression $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	

Students will be assessed on their ability to:	Suggested experiments
113 use the expression $pV = NkT$ as the equation of state for an ideal gas	Use temperature and pressure sensors to investigate relationship between p and T Experimental investigation of relationship between p and V
114 show an awareness of the existence and origin of background radiation, past and present	Measure background count rate
115 investigate and recognise nuclear radiations (alpha, beta and gamma) from their penetrating power and ionising ability	Investigate the absorption of radiation by paper, aluminium and lead (radiation penetration simulation software is a viable alternative)
116 describe the spontaneous and random nature of nuclear decay	
117 determine the half lives of radioactive isotopes graphically and recognise and use the expressions for radioactive decay: $dN/dt = -\lambda N$, $\lambda = \ln 2/t_{½}$ and $N = N_0 e^{-\lambda t}$	Measure the activity of a radioactive source Simulation of radioactive decay using, for example, dice
136 explain the concept of nuclear binding energy, and recognise and use the expression $\Delta E = c^2 \Delta m$ and use the non SI atomic mass unit (u) in calculations of nuclear mass (including mass deficit) and energy	
137 describe the processes of nuclear fusion and fission	
138 explain the mechanism of nuclear fusion and the need for high densities of matter and high temperatures to bring it about and maintain it	
118 discuss the applications of radioactive materials, including ethical and environmental issues	
126 use the expression $F = Gm_1m_2/r^2$	
127 derive and use the expression $g = -Gm/r^2$ for the gravitational field due to a point mass	
128 recall similarities and differences between electric and gravitational fields	
129 recognise and use the expression relating flux, luminosity and distance $F = L/4\pi d^2$	application to standard candles

Students will be assessed on their ability to:	Suggested experiments
130 explain how distances can be determined using trigonometric parallax and by measurements on radiation flux received from objects of known luminosity (standard candles)	
131 recognise and use a simple Hertzsprung-Russell diagram to relate luminosity and temperature. Use this diagram to explain the life cycle of stars	
132 recognise and use the expression $L = \sigma T^4 \times$ surface area, (for a sphere $L = 4\pi r^2 \sigma T^4$) (Stefan-Boltzmann law) for black body radiators	
133 recognise and use the expression: $\lambda_{max} T = 2.898 \times 10^{-3}$ m K (Wien's law) for black body radiators	
134 recognise and use the expressions $z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source of electromagnetic radiation moving relative to an observer and $v = H_o d$ for objects at cosmological distances	
135 be aware of the controversy over the age and ultimate fate of the universe associated with the value of the Hubble Constant and the possible existence of dark matter	